

SGA 2: Application to the Fundamental Group

Bounded source-aligned working unit — 20 July 2026

Authority. Corrected French TeX lines 3425–3427; original printed p. 116, physical source-PDF p. 101, and recomposed running p. 93. These locator systems are kept distinct. Blank line 3428 is excluded; the raw cursor is line 3428 and the next substantive cursor is line 3429 (Corollary 2.4). The complete proof, all covering types, hats, primes, references, and conclusion were checked directly. The external English candidate is comparison-only.

Proof. Part (i) has already been proved.

Part (ii) follows from (i) and the hypothesis, at least for L_U and P_U . Moreover, if R is an étale covering of $\widehat{X}$, there are an open neighborhood U of Y in X and a finite flat covering R' of U such that

$$\widehat{R'} \simeq R.$$

By restriction this gives a covering R'' of Y , which is étale by Proposition 1.1. Hence the restriction of R' to some neighborhood U' of Y is étale. $\square$